SQL BASIC

Q . Find the 5 cheapest products, showing the lowest price first.

- Select name and price
- Sort by price in ascending order (lowest first)
- Limit to 5 results

Query :

SELECT name, price FROM products

ORDER BY price ASC

LIMIT 5;

Q. Find the top 3 highest-rated products.

- Select name, price, and rating
- Sort by rating in descending order (highest first)
- Limit to 3 results

Query :

SELECT name, price, rating FROM products

ORDER BY rating DESC

LIMIT 3;

Q. Find the 3 most expensive products in the Electronics category (category_id = 1).

- Select name and price
- Filter for category_id = 1 (Electronics)
- Sort by price descending (most expensive first)
- Limit to 3 results

Query :

SELECT name, price FROM products

WHERE category_id = 1

ORDER BY price DESC

LIMIT 3;

Q. Count the total number of products in the store.

- Use COUNT(*) to count all rows
- Return a single number

Query :

SELECT Count(*) FROM products;

Q. Calculate the minimum, maximum, and average price of all products.

- Calculate MIN(price) as min_price
- Calculate MAX(price) as max_price
- Calculate AVG(price) as avg_price

Return columns: min_price, max_price, avg_price

Query :

SELECT MIN(price) as 'min_price',

MAX(price) as 'max_price',

AVG(price) as 'avg_price'

from products;

Q. Calculate the total revenue from all orders.

Query :

SELECT SUM(total) as 'total_revenue' FROM orders;

Q. Count how many products are in each category.

Query :

SELECT category_id, count(*) as 'product_count' FROM products

GROUP BY category_id;

Q. Find the average price of products in each category.

Query :

SELECT category_id, AVG(price) as 'avg_price'

FROM products

GROUP BY category_id;

Q. Find customers who have spent more than ₹50,000 total.

Query :

```
SELECT customer_id, SUM(total) as 'total_spend'
FROM orders
GROUP BY customer_id
HAVING SUM(total) > 50000
ORDER BY total_spend DESC;
```

Q. Get product names along with their category names.

Query :

```
SELECT p.name, c.name
FROM products p INNER JOIN categories c
on p.category_id = c.id;
```

Write a query to display product id, brand, product category and color of all clothing type products.

Query :

```
SELECT ProductId, Brand, PCategory, Color
FROM Product
WHERE PCategory = 'Clothing';
```

Problem Statement

Write a query to display vendorid, vendorname and rating of those vendors that do not have any rating.

Query :

```
SELECT VendorId, VendorName, Rating
FROM Vendor
WHERE Rating is NULL
```

Problem Statement

Write a query to display CustId, CustName, Age and Email of those customers who have ‘a’ in the CustName at any position and ‘u’ in the CustLocation at any position.

Query :

SELECT CustId, CustName, Age, Email

FROM Customer

Where CustName LIKE '%a%' AND CustLocation LIKE '%u%'

Problem Statement

Identify the journals from ‘Scientific’ or ‘Politics’ genre. Display the journalid and journalname of those journals.

Query :

SELECT JournalId, JournalName

FROM Journal

WHERE Genre = 'Scientific' or Genre = 'Politics'

Problem Statement

The publishers want to give a discount of 5% on monthly charges. Write a query to display the following:

Query :

SELECT JournalName,

MonthlyCharges as "CURRENT CHARGES",

*MonthlyCharges * (1-5/100) as "OFFER PRICE",*

*MonthlyCharges * (5/100) as "DISCOUNTED AMOUNT"*

FROM Journal

Problem Statement

Retrieve subscriptionid and customerid of the customers who have their startdate between the dates '1-Jan-2023' and '30-Jun-2023';

Query :

SELECT subscriptionid, customerid

FROM subscription

WHERE StartDate between '2023-01-01' AND '2023-06-30'

Problem Statement

Display customerid, customername, state and city of customers whose type is other than Regular .

Query :

SELECT Customerid, Customername, state, city

FROM Customer

WHERE Customertype <> 'Regular'

Problem Statement

Retrieve the journal name and genre of journals whose genre is either 'Trade' or 'Politics' and monthly charges lie in the range of 90 to 125 (both inclusive).

Query :

SELECT JournalName, Genre

FROM Journal

WHERE (Genre = 'Trade' or Genre = 'Politics') AND

(Monthlycharges BETWEEN 90 AND 125)

Problem Statement

Display CustomerId, CustomerName, CustomerType and State of the customer(s) from Illinois who have 'rime' or 'gul' in their customertype.

Query :

SELECT Customerid, customername, customertype, state

FROM Customer

WHERE (state = 'Illinois') AND

(customertype LIKE '%rime%' OR customertype LIKE '%gul%')

Problem Statement

Retrieve the publishername of publishers who do not have contact number.

Query :

SELECT Publishername

FROM Publisher

WHERE contactno is NULL

Problem Statement

Retrieve journal id, journal name and monthly charges for journals that have monthly charges in the range 85 to 130 (excluding these values).

Query :

SELECT JournalId, JournalName, MonthlyCharges

FROM Journal

WHERE MonthlyCharges BETWEEN 86 AND 129

Problem Statement

Retrieve the journalid and monthlycharges of all 'Trade' and 'Politics' genre.

Query :

SELECT journalId, monthlyCharges

FROM journal

WHERE (Genre = 'Trade' AND Genre = 'Politics')

Problem Statement

Display the names of states in which the customers reside.

Query :

SELECT Distinct state

FROM Customer

Problem Statement

Retrieve all the unique dates on which any subscription has been started.

Query :

SELECT Distinct StartDate

FROM Subscription

Problem Statement

For each customer type display the states in which customers reside.

Query:

SELECT Distinct CustomerType, State

FROM Customer

Problem Statement

Display JournalName and Genre of journal(s) whose Genre contains the string 'ti'.

Query :

Select journalName, Genre

FROM Journal

WHERE Genre LIKE '%ti%'

Problem Statement

Retrieve publisher details for publishers whose names have 'i' as the second last character.

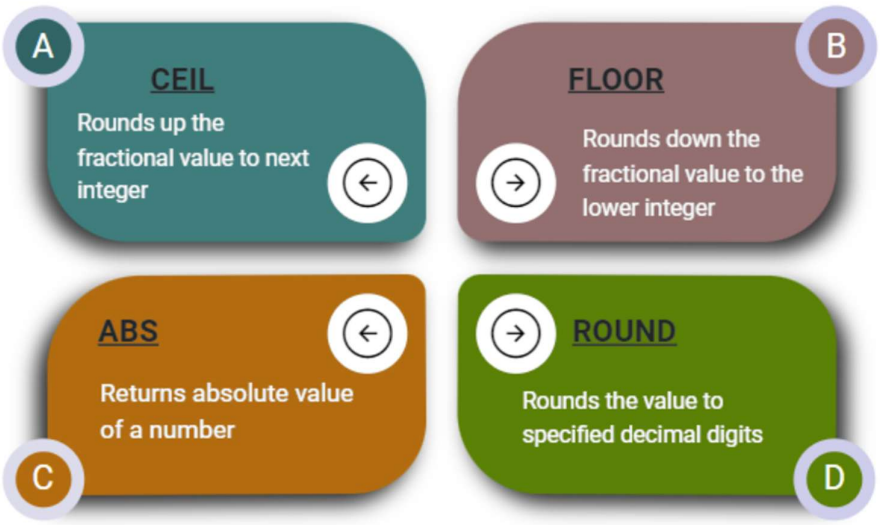
Query :

SELECT PublisherId, PublisherName, Rating, ContactNo

FROM Publisher

WHERE PublisherName LIKE '%i_'

Numeric Functions



Problem Statement

For every product offered by the vendor, display the VendorId as "VendorId", price values rounded off to the nearest higher and lower whole numbers (integers) as "Higher_Round" and "Lower_Round" for every product-vendor combination.

Query :

SELECT vendorId as "VendorId",

CEIL(Price) as "Higher_Round",

FLOOR(Price) as "Lower_Round"

FROM Vendoroffering

Problem Statement

Display the price of each product offered by the vendor after applying a discount of 500 rupees as "After Discount". The price should be displayed as positive values.

Query :

SELECT ABS(Price - 500) as "After Discount"

FROM VendorOffering

CHARACTER FUNCTIONS :

UPPER, LOWER, CONCAT, LENGTH

Problem Statement

Identify all white colored products whose product name has more than 9 characters. Display numeric part of ProductId and PName separated by hyphen (-) for the identified products. (Do case-insensitive comparison). For the given sample data, the following is the expected output.

Query :

SELECT CONCAT(SUBSTRING(ProductId, 2), '-', PName) as "Product"

FROM Product

WHERE LOWER(Color) = 'white' AND LENGTH(Pname) > 9

Problem Statement

Display the product name along with the first character of its review seperated by hyphen ('-') for those products whose category contains the character 'i' anywhere. Display it as "Product_Review".

Query :

SELECT CONCAT(Pname, '-', SUBSTR(Review, 1, 1)) as "Product_Review"

FROM Product

WHERE PCategory LIKE '%i%'

Problem Statement

Identify the vendors whose rating is known. For the identified vendors, display the minimum rating as "MIN_RATING", maximum rating as "MAX_RATING", average rating as "AVG_RATING "and number of vendors as "NUMBEROFVENDORS".

Query :

SELECT MIN(Rating) as "MIN_RATING",

MAX(Rating) as "MAX_RATING",

AVG(Rating) as "AVG_RATING",

COUNT(*) as "NUMBEROFVENDORS"

FROM Vendor

WHERE RATING is NOT NULL

Problem Statement

Display Maximum age, Minimum age, Average age and count of those customers whose name contains more than 3 characters. The display should have column names as "MaxAge", "MinAge" , "AvgAge" and "NumberOfCustomers".

Query :

SELECT MAX(Age) as "MaxAge",

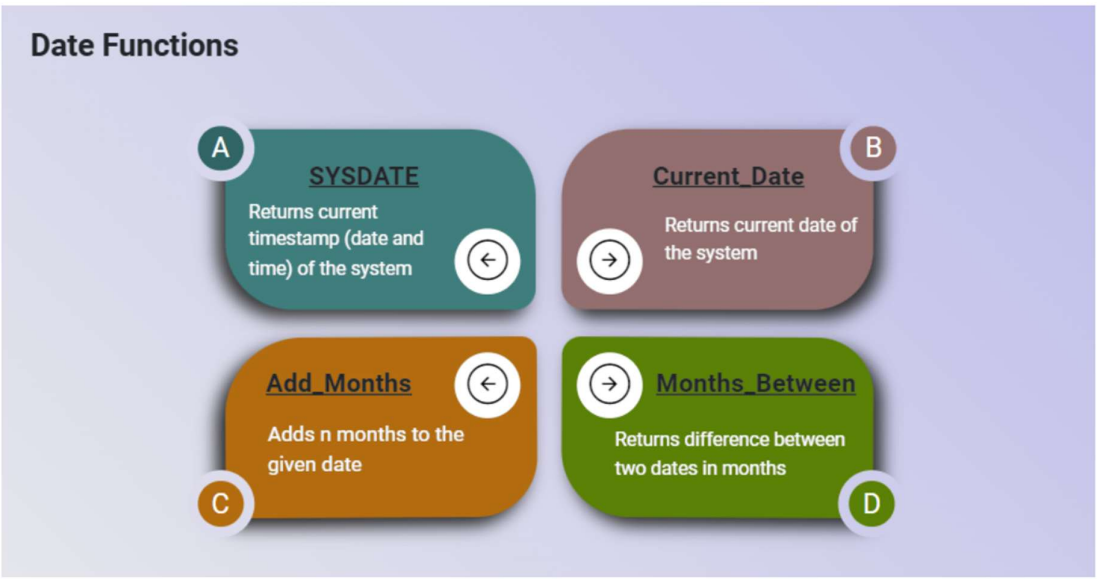
MIN(Age) as "MinAge",

AVG(Age) as "AvgAge",

COUNT(*) as "NumberOfCustomers"

FROM Customer

WHERE LENGTH(CustName) > 3



Problem Statement

Display invoice id, product id and months lapsed betweeen current date and purchase date for the product 'P105'.

Query :

SELECT InvoiceId, productId,

ROUND(MONTHS_BETWEEN((SELECT CURRENT_DATE FROM Information_Schema.System_users), PurchaseDate)) as "NOOFMONTHS"

FROM Invoice

WHERE productId = 'P105'

Problem Statement

Assume that in the supermarket scenario, every product has an exchange validity of 2 months, starting from the date of purchase.
Display the Invoiceid as "InvoiceId", PurchaseDate as "PurchaseDate" and exchange validity date as "ExchangeValidityDate" of all the sold products.

Query :

SELECT InvoiceId as "InvoiceId",

PurchaseDate as "PurchaseDate",

ADD_MONTHS(PurchaseDate, 2) as "ExchangeValidityDate"

FROM Invoice

NVL – Can be used to fill out the null values

Problem Statement

Display the CustId as "CustomerId", CustName as "CustomerName", Email as "CustEmail" and Age as "CustAge" of all customers. If the emailid and age are unknown, display "Not Provided" and 0 respectively.

Query :

SELECT CustId as "CustomerId",

CustName as "CustomerName",

NVL(Email, 'Not Provided') as "CustEmail",

NVL(Age, 0) as "CustAge"

FROM Customer

Problem Statement

Display VendorId and Rating as "RATING" of all the vendors. If the rating is unknown, display "Not Provided".

Query:

SELECT VendorId as "VENDORID",

NVL(TO_CHAR(Rating), 'Not Provided') as "RATING"

FROM Vendor

CASE : can add the values to the column depending upon the condition

```
CASE <column_name>
  WHEN expression1 THEN result1
  WHEN expression2 THEN result2
  ...
 ELSE resultN
END
```

```
1. CASE Designation
2.   WHEN 'SE' THEN Salary * 1.2
3.   WHEN 'SSE' THEN Salary * 1.1
4.   ELSE Salary * 1.05
5. END
```

PRACTICE PROBLEMS :

Problem Statement

Apply a discount of 70.25 rupees on the MonthlyCharges for those journals whose Genre contains more than 8 characters. Display the JournalId as "JournalId" and round off the calculated price to the lower(integer) and display as "Rounded_Integer".

Query :

SELECT JournalId as "JournalId",

ROUND(FLOOR(MonthlyCharges - 70.25)) as "Rounded_Integer"

FROM Journal

WHERE LENGTH(Genre) > 8

Problem Statement

Identify the publishers whose contact number is known and their name contains more than 5 characters. Display the number of such publishers as "NUMBEROFPUBLISHERS".

Query :

SELECT COUNT(*) as "NUMBEROFPUBLISHERS"

FROM Publisher

WHERE ContactNo is NOT NULL AND LENGTH(PublisherName) > 5

Problem Statement

The monthly charges of 'Academics' Journals is reduced by 37.5%. Display the JournalId as "JournalId", Genre as "Genre", existing monthlycharges as "Old_Price", and revised monthlycharges as "New_Price" for all the 'Academic' Journals. Revised Monthly charges should be rounded up to 2 decimal places.

Query :

```
SELECT JournalId as "JournalId",
Genre as "Genre",
MonthlyCharges as "Old_Price",
ROUND((MonthlyCharges * 0.625), 2) as "New_Price"
From Journal
WHERE Genre = 'Academics'
```

Problem Statement

Display CustomerId and CustomerName of 'Regular' customer(s) who reside in a state that contains character 'o' anywhere.
(Do case-insensitive comparison for CustomerType)

Query :

```
SELECT CustomerId,
CustomerName
FROM Customer
WHERE LOWER(CustomerType) = 'regular' AND State LIKE '%o%'
```

Problem Statement

Display JournalId and JournalName for all the journal(s) whose last digit of the publisher id is more than 2 and journal name contains character 'o' and 't' anywhere, but in the same order.

Query:

SELECT JournalId,

JournalName

FROM Journal

WHERE SUBSTR(publisherId, LENGTH(publisherId), 1) > 2 AND

JournalName LIKE ('%o%t%')

Problem Statement

Display Maximum, Minimum and Average values of DiscountPercent which is offered on any subscription. The display should have the column names as "MaxDiscount", "MinDiscount" and "AvgDiscount".

Query :

SELECT MAX(DiscountPercent),

MIN(DiscountPercent),

AVG(DiscountPercent)

FROM Subscription

Problem Statement

Display CustomerId along with the first four character of their CustomerType separated by slash ('/') as "Customer-ID".

Query :

SELECT CONCAT(CustomerId, '/', SUBSTR(CustomerType, 1, 4)) as "Customer-ID"

FROM Customer

Format Date

Parts of Date

To fetch the parts of date like date, month and year.

```
1. SELECT TO_CHAR(recorddate,'DD') "DATE",
   TO_CHAR(RecordDate,'MM') "MONTH",
   TO_CHAR(recorddate,'YY') "YEAR" from Weather;
```

Output:

Problem Statement

Display the total number of unique StartDates as "Total_Dates" on which the subscription is taken.

Query :

SELECT Count(DISTINCT((StartDate))) as "Total_Dates"

From Subscription

Problem Statement

For the Subscriptions bought in the month of November, display the minimum discount(percent) offered as "Min Discount Percent", along with the total number of customers who subscribed in that month as "No. Of Customers"

Query :

SELECT MIN(DiscountPercent) as "Min Discount Percent",

COUNT(*) as "No. Of Customers"

FROM Subscription WHERE TO_CHAR(StartDate, 'MM') = 11

Problem Statement

Find the number of unique customers who have subscribed to a journal in any month except 'June'. Display the identified number of unique customers as "NoOfUniqueCustomer".

Query :

```
SELECT COUNT(DISTINCT(CustomerId)) as "NoOfUniqueCustomer"
FROM Subscription
WHERE TO_CHAR(StartDate, 'MM') <> 6
```

Problem Statement

Display the total MonthlyCharges as "Monthly_Charges" of all the Journals whose last digit of the numeric part of PublisherId is more than 4.

Query :

```
SELECT SUM(MonthlyCharges) as "Monthly_Charges"
FROM Journal
WHERE SUBSTR(PublisherId, LENGTH(PublisherId), 1) > 4
```

Problem Statement

On 30th June 2024, an executive is asked to send subscription renewal reminder to the customers whose subscription is going to end within a month. For this, he wants to identify the subscription id and corresponding customer id.

Query :

```
SELECT SubscriptionId,
      CustomerId
FROM Subscription
WHERE DATEADD(month, DurationinMonths, StartDate)
     BETWEEN '2024-06-30' AND '2024-07-30';
```

Problem Statement

A publisher wants to categorize the journals based on the monthly charges of the journals. Write a query to display JournalName, Monthly Charges and JournalCategory. The categorization is done as: If the MonthlyCharges is less than 75, then the JournalCategory will be 'LowCost'. Else if, MonthlyCharges is between 75 and 100, then the JournalCategory will be 'AverageCost'. Else, the JournalCategory will be 'HighCost'.

Query :

```
SELECT JournalName,
MonthlyCharges,
Case
   WHEN MonthlyCharges < 75 THEN 'LowCost'
   WHEN MonthlyCharges BETWEEN 75 AND 100 THEN 'AverageCost'
   ELSE 'High Cost'
END as "JournalCategory"
FROM Journal
```

Problem Statement

Monthly charge of Journal is appraised to certain percentage based on the genre as per the table given below. Find the new monthly charges for all the journals. Write a query to display JournalName, Genre, MonthlyCharges and NewMonthlyCharges, arranged in ascending order of Genre.

GENRE	Appraisal %
Trade	10
Scientific	15
Academics	5
Politics	25
Others	20

Query :

```
SELECT JournalName as "JOURNALNAME",
Genre as "GENRE",
MonthlyCharges as "MONTHLYCHARGES",
Case
   WHEN Genre = 'Trade' THEN MonthlyCharges + (MonthlyCharges * 0.1)
   WHEN Genre = 'Scientific' THEN MonthlyCharges + MonthlyCharges * 0.15
   WHEN Genre = 'Academics' THEN MonthlyCharges + MonthlyCharges * 0.05
   WHEN Genre = 'Politics' THEN MonthlyCharges + MonthlyCharges * 0.25
   WHEN Genre = 'Others' THEN MonthlyCharges + MonthlyCharges * 0.2
END as "NEWMONTHLYCHARGE"
FROM Journal
ORDER BY Genre
```

Query :

Problem Statement

Display product id, product category and brand of the products which belongs to the category of 'Clothing' or 'Accessories' in ascending order of the product category and review.

Query :

```
SELECT ProductId,
Pcategory,
Brand
FROM Product
WHERE Pcategory in ('Clothing', 'Accessories')
ORDER BY Pcategory ASC
```

Problem Statement

Display the product id, product category, brand and color of all products in descending order of product category and ascending order of color.

Query :

SELECT ProductId,

Pcategory,

Brand,

Color

FROM Product

ORDER BY Pcategory DESC, Color ASC

Problem Statement

Display PublisherId, PublisherName and Rating of all the publishers in ascending order of Rating and PublisherName.

Query :

SELECT PublisherId, PublisherName, Rating

FROM Publisher

ORDER BY Rating ASC, PublisherName ASC

Problem Statement

Display the SubscriptionId, DurationInMonths and DiscountPercent of the Subscription(s) for which the DurationInMonths is more than 6 and arrange them in increasing order of DurationInMonths and decreasing order of StartDate.

Query :

SELECT SubscriptionId,

DurationInMonths,

DiscountPercent

FROM Subscription

WHERE DurationInMonths > 6 ORDER BY DurationInMonths ASC, StartDate DESC